

Treatment of Na_v1.7-mediated pain in inherited erythromelalgia using a novel sodium channel blocker

Yigal Paul Goldberg^{a,*}, Nicola Price^a, Rostam Namdari^a, Charles Jay Cohen^a, Mieke H. Lamers^b, Conrad Winters^a, James Price^c, Clint E. Young^a, Henry Verschoof^a, Robin Sherrington^a, Simon Neil Pimstone^a, Michael Reuben Hayden^{a,d}

^a Xenon Pharmaceuticals Inc., Burnaby, British Columbia, Canada

^b Radboud University Nijmegen Medical Center, Nijmegen, The Netherlands

^c Strength in Numbers Mathematical and Statistical Services, Vancouver, British Columbia, Canada

^d University of British Columbia, Department of Medical Genetics, Center for Molecular Medicine and Therapeutics, Vancouver, British Columbia, Canada

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ABSTRACT

Mutations in the SCN9A gene leading to deficiency of its protein product, Na_v1.7, cause congenital indifference to pain (CIP). CIP is characterized by the absence of the ability to sense pain associated with noxious stimuli. In contrast, the opposite phenotype to CIP, inherited erythromelalgia (IEM), is a disorder of spontaneous pain caused by missense mutations resulting in gain-of-function in Na_v1.7 that promote neuronal hyperexcitability. The primary aim of this study was to demonstrate that Na_v1.7 antagonism could alleviate the pain of IEM, thereby demonstrating the utility of this opposite phenotype model as a tool for rapid proof-of-concept for novel analgesics. An exploratory, randomized, double-blind, 2-period crossover study was conducted in 4 SCN9A mutation-proven IEM patients. In each treatment period (2 days), separated by a 2-day washout period, patients were orally administered XEN402 (400 mg twice daily) or matching placebo. In 3 patients, pain was induced by heat or exercise during each treatment arm. A fourth patient, in constant severe pain, required no induction. Patient-reported outcomes of pain intensity and/or relief were recorded, and the time taken to induce pain was measured. The ability to induce pain in IEM patients was significantly attenuated by XEN402 compared with placebo. XEN402 increased the time to maximal pain induction and significantly reduced the amount of pain (42% less) after induction ($P = .014$). This pilot study showed that XEN402 blocks Na_v1.7-mediated pain associated with IEM, thereby demonstrating target engagement in humans and underscoring the use of rare genetic disorders with mutant target channels as a novel approach to rapid proof-of-concept.

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1. Introduction

Drugs targeting novel analgesic mechanisms that appeared promising in the preclinical setting have often failed to translate that promise during clinical development. These failures were, to a large degree, due to inadequate human validation of the target. Here we present a novel application of human genetics, illustrating the important role that rare genetic disorders can play in the development of novel drugs that may be of relevance for a common indication, such as for patients with the more common forms of neuropathic pain.

Congenital indifference to pain (CIP) is a rare autosomal recessive disorder characterized by the complete absence of pain sensation typically associated with noxious stimuli [3,4,11,12,14]. For example, patients have reported pain-free episodes of childbirth, corneal abrasions, fractures, and severe burns [3,11]. CIP is caused by a genetic deficiency of the SCN9A gene encoding the voltage-gated sodium channel, Na_v1.7 [2,3,11,14]. This finding underscores the essential role of Na_v1.7 in mediating pain signaling in humans, and suggests that mimicking the CIP phenotype by Na_v1.7 antagonism may be a highly effective strategy to treat human painful conditions [3,10,11].

In contrast, inherited erythromelalgia (IEM) is a rare autosomal dominant condition, characterized by debilitating spontaneous or easily evoked attacks of symmetrical burning pain in the feet and hands, typically associated with elevated skin temperature and erythema [2,14]. Symptoms are generally induced by exercise,

* Corresponding author. Address: Xenon Pharmaceuticals Inc., 3650 Gilmore Way, Burnaby, BC, Canada V5G 4W8. Tel.: +1 604 484 3362; fax: +1 604 484 3321.
E-mail address: pgoldberg@xenon-pharma.com (Y.P. Goldberg).

prolonged standing, exposure to heat, and/or changes in humidity [6]. The pain can be so severe that it can lead to suicide, and adequate treatment remains very challenging. IEM is caused by SCN9A missense mutations resulting in a gain-of-function in the Na_v1.7 channel promoting hyperexcitability of dorsal root ganglion neurons, which results in aberrant pain signaling in the absence of a noxious stimulus [5,14,15]. The spontaneous and severe pain observed in this disorder represents the opposite of the absence of pain observed in CIP. Thus, the antagonism of Na_v1.7, in an effort to mimic the CIP Na_v1.7-loss-of-function and analgesic phenotype, represents a rational treatment for IEM, as it should inhibit the underlying hyperexcitable Na_v1.7 channel responsible for the debilitating pain [9,11]. Furthermore, gain-of-function of Na_v1.7 is likely to be a common component of other painful conditions because a SNP present in approximately 10% of the general population is associated with sensitization to pain [13].

We have developed a novel compound (XEN402) that exhibits potent, voltage-dependent block of Na_v1.7 (IC₅₀ 80 nM). Here we report an exploratory trial performed to evaluate the efficacy of XEN402 in mutation-proven erythromelalgia patients, with the primary aim of demonstrating that Na_v1.7 antagonism alleviates pain associated with IEM. Our novel approach uses a rare genetic disease with extreme phenotypic characteristics to identify a novel target for drug development [11], and to provide preliminary clinical evidence for pharmacological modulation of the target (ie, target engagement) by a novel drug. This approach represents an alternative to larger, heterogeneous proof-of-concept trials, enabling more rapid, cost-effective and mechanistically highly relevant clinical data for optimization of novel drugs to treat common disorders, such as neuropathic pain.

2. Methods

2.1. Trial design

This single-centre, multiple-dose, in-patient, randomized, double-blind, placebo-controlled, crossover trial was conducted between April 12 and July 7, 2010, at Radboud University Medical Center's Clinical Research Centre Nijmegen, in the Netherlands (Supplemental Fig. 1). This trial is registered with ClinicalTrials.gov, number NCT01090622.

Patients 18 to 75 years of age with a confirmed clinical and molecular diagnosis of erythromelalgia were eligible to participate in this study. Patients were excluded if they had significant other illnesses or were unable to refrain from using all concomitant medications while resident in the clinic.

Subjects were required to report a sufficient level of either spontaneous or induced pain to be randomized (≥ 4 on the Pain Intensity Numerical Rating Scale [PINRS], and/or moderate/severe on the Categorical Rating Scale [CRS], or an intolerable level of pain irrespective of pain scores). If a subject was unable to spontaneously report "qualifying pain," then pain was induced to determine whether that individual qualified for the trial, and pain was subsequently induced at various time points after dosing. Subjects were allocated to different groups that were evaluated and analyzed independently depending on their requirement to induce pain: 1 subject was allocated to Group A (no pain induction required) and 3 subjects to Group B (pain induction required). Four subjects received XEN402 400 mg twice daily or matching placebo for 2 consecutive days in each treatment period. The 2 treatment periods were separated by a 2-day washout period.

Use of all medications (except rescue) was prohibited from 2 days before dosing until discharge. Two subjects stopped using medication to participate in the trial. Subject 1001 took her final dose of methadone 5 days before dosing. Subject 1003 took his

final doses of Diovan and Maxalt 3 days before dosing. If required, nonpharmacological rescue measures (cooling, rest, elevation) were permitted to alleviate pain. If these methods proved ineffective, 500 mg acetaminophen at 4- to 6-hour intervals up to a maximum of 3 g per day was permitted as rescue medication.

The study was approved by the investigative site's research ethics committee (Committee on Research Involving Human Subjects Arnhem-Nijmegen Region). The study was performed in accordance with the ethical principles of the Declaration of Helsinki, and was consistent with good clinical practice guidelines and applicable regulatory requirements. Written informed consent was obtained from all subjects before initiating study-related procedures.

2.2. Randomization and masking

As subjects qualified for dosing they were randomly allocated to 1 of 2 double-blind treatment sequences: XPF-001/Placebo or Placebo/XPF-001. To maintain blinding, XPF-001 and placebo capsules were identical in size and appearance.

2.3. Interventions and outcome assessments

All 4 subjects scored their pain using an 11-point PINRS (0 = no pain to 10 = worst pain imaginable), and a 4-point CRS of pain severity (none, mild, moderate, severe) [7,8,11]. For the 3 subjects in Group B (subjects determined not to have spontaneous moderate levels of pain without induction), pain was induced 6 times during each treatment period (2 and 6 hours after each morning dose and 4 hours after each evening dose) using the same individually standardized method defined before randomization. Heating of subjects' hands or feet using an electric heater placed at a standardized distance from the hands or feet was the preferred method of induction, but subjects could also use exercise to induce pain. During induction, subjects were required to record their pain scores (PINRS and CRS) at 15-minute intervals. Induction was stopped either when subjects achieved a score of at least 4 on the PINRS or 'moderate' on the CRS scale, an intolerable level of pain was achieved, or when the investigator determined that the patient was refractory to the stimulus and that further heating would have no impact. Subjects continued to rate their pain at regular intervals for up to 4 hours after completion of each induction. The subject in Group A (for whom pain was not induced) was required to record her pain scores (PINRS and CRS) at regular intervals for up to 8 hours after each dose administration. To assess the degree of pain relief, the subject in Group A was also asked to complete a 5-point Pain Relief Categorical Rating Scale (REL) (none, a little, some, a lot, complete) at regular intervals for up to 8 hours after each dose.

Blood and urine samples for laboratory assessments (serum chemistry, hematology, coagulation, and urinalysis) were collected at the following times during each treatment period: before each morning dose, 2 hours after the first dose on Day 1, and 4 hours after the evening dose on Day 2. ECGs were performed at the following times during each treatment period: before the morning dose, 2 hours after the morning dose, and 4 hours after the evening dose. Vital signs were performed at the following timepoints during each treatment period: before and at 1 and 4 hours after every dose. Adverse events (AEs) were recorded at the time of occurrence from Screening until follow-up. Blood samples for pharmacokinetic analysis were collected at multiple timepoints during each treatment period. These data are not presented.

2.4. Statistical methods

This was an exploratory study without prespecified efficacy endpoints. Because of the small number of subjects enrolled,

graphical presentation was used to elucidate within-subject drug effects and/or overall findings. Where sufficient replicate data were available, exploratory statistical analyses were performed using simple linear models using treatment and subject as fixed and random effects where appropriate.

For subjects in Group B, as a measure of the amount of pain after induction, the AUC for PINRS for the 2 hours after each induction was calculated using the linear trapezoidal rule. Missing data were imputed using a 'last-observation-carried-forward' algorithm in both directions. For the subject in Group A, Total Pain Relief (TOTPAR) was defined as the time-weighted sums of the relief scores over 4 or 8 hours after each morning study drug administration, and sums of relief scores over 6 hours after each evening study drug administration, respectively.

2.5. Study population

2.5.1. Subject 1001

Subject 1001 is a 24-year-old woman with severe pain in both feet. Her symptoms began 8 months before the trial with tingling in both feet, but she subsequently developed severe pain (PINRS 9) in both her hands and feet, which is associated with redness of her feet. The pain is maximal even at rest, especially in the evening and during the nights, disturbing her sleep. Alcohol aggravates the pain. Her medical and family histories were negative. Neurological evaluation including nerve conduction studies was unremarkable. The patient had been treated with pregabalin and methadone with minimal effect. SCN9A sequencing identified a p.R185H mutation.

2.5.2. Subject 1002

Subject 1002 is a 39-year-old man who presented in late 2008 with burning pain in the feet and lower legs followed by burning of the distal arms. He also experienced numbness in the plantar side of first digit in both feet. The patient reported areas of hyperhidrosis without skin discoloration. His family history was unremarkable. Physical examination was normal except for reduced light-touch and pain sensation at the plantar side of the big toes.

Nerve conduction studies and cerebral magnetic resonance imaging examination were normal. SCN9A sequencing identified a p.I720K mutation.

2.5.3. Subject 1003

Subject 1003, a 35-year old man, presented with a history of cramps, warmth and redness particularly of the upper extremities. At the age of 10 years, he experienced episodes of redness of the skin in his hands and wrists that lasted for hours. These symptoms were mainly induced by warmth and exercise, and relieved by cold water immersion. In the past decade, the frequency and duration of the episodes increased and the skin redness had extended to the feet, cheeks, and ears, and become painful. Treatment with aspirin and a β -blocker were ineffective. His father and brother reported similar complaints of redness of the skin, but without pain. Physical examination revealed red and swollen hands with eczema and pes cavus. SCN9A sequencing identified a p.G856D mutation.

2.5.4. Subject 1004

Patient 1004 is a 46-year-old man with pain in both feet and hands, which started when he was 5 years old. His complaints were burning, painful, swollen, red, warm feet and hands. His feet were more affected than his hands. The symptoms were mainly induced by altered humidity, warmth and exercise. The only medical therapy he used in the past was codeine which proved ineffective. His medical history was unremarkable. Family history revealed a mother and 2 maternal aunts with similar symptoms, of variable severity. Physical examination did not show any abnormalities except for redness of the skin at the medial aspect of his big toes. SCN9A sequencing identified a p.F1449V mutation.

2.6. Role of the funding source

The trial sponsor designed the trial, collected and interpreted the data and wrote this report. Statistical analyses were performed by an independent statistical services company. All data were reviewed by 1 of the authors (Y.P.G.).

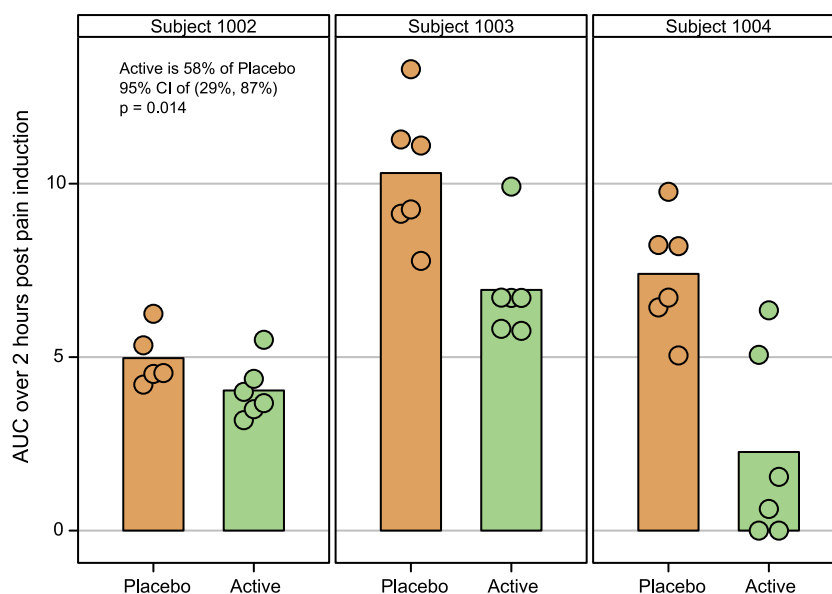


Fig. 1. Reduction in total pain in the 2 hours after each pain induction by treatment for subjects 1002, 1003, and 1004. Pain was induced at 2 and 6 hours after the morning (AM) dose and 4 hours after the evening (PM) dose on each dosing day, for a total of 6 induction periods per treatment. The amount of pain for the 2 hours after each induction was quantified using AUC₀₋₂ PINRS scores. For subject 1002, the AUC for the 2 hours after induction is 79% of placebo (95% CI = 67%, 92%; $P = .011$), whereas for subject 1003, the AUC for the 2 hours after induction is 67% of placebo (95% CI = 55%, 83%; $P = .004$), and for subject 1004, the AUC for the 2 hours after induction is 12% of placebo (95% CI = 2%, 75%; $P = .031$).

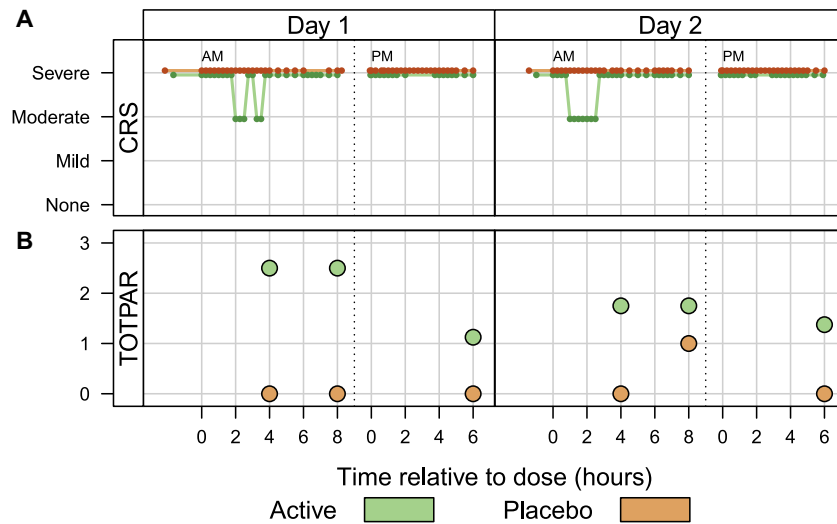


Fig. 2. Pain relief for subject 1001. (A) CRS scores for subject 1001. The CRS was reported by the subject throughout the day and each score is represented by an individual dot. CRS = categorical rating scale in which pain is rated as none, mild, moderate, or severe. (B) TOTPAR-4, -6, and -8 for subject 1001. Total pain relief (TOTPAR) was calculated at 4 hours (TOTPAR-4) and 8 hours (TOTPAR-8) after the morning doses, and TOTPAR-6 was calculated 6 hours after the evening doses.

3. Results

Five subjects provided informed consent and were screened for this study. One subject did not fulfill the eligibility criteria, and 4 subjects were subsequently randomized to 1 of 2 treatment sequences. All 4 subjects were dosed. No subject discontinued from the study. Data for all 4 subjects were included in the safety and efficacy analyses. Subject 1001 was experiencing chronic severe pain, and no attempt was made to further induce her pain. The other 3 subjects presented with episodic spontaneous pain and pain induction procedures were used to facilitate measurements of efficacy. Efficacy and safety results are reported here; pharmacokinetic data are not reported.

3.1. Efficacy outcomes

Overall, the results indicate that there were improvements in pain efficacy measures in all 4 subjects with statistically significant reductions in pain scores in the 3 subjects in whom pain was induced. The amount of pain in the 2 hours after induction was reduced by 21% ($P = .011$), 33% ($P = .004$), and 88% ($P = .031$) in these 3 patients, respectively (Fig. 1). Overall, in these 3 subjects, pain was reduced by 42% on XEN402 compared with placebo (95% CI = 13%, 71%; $P = .014$). Subject 1001 was in constant pain, so the focus of her assessments was on measuring pain relief. Subject 1001 showed improvements in her CRS and TOTPAR scores (Fig. 2).

3.2. Subject 1001

Subject 1001 (in constant pain) reported an improvement in pain measured using the CRS on XEN402 (Fig. 2A), with corresponding improved pain relief scores (data not shown). Her total pain relief scores at 4, 6, and 8 hours (TOTPAR-4, -6, -8) showed improvement over placebo (Fig. 2B).

3.3. Subject 1002

In subject 1002, pain was not successfully induced using exercise, resulting in a relatively small elevation in pain scores, hindering the ability to observe a robust analgesic response to XEN402 treatment (Fig. 3A). Despite this lack of pain induction, subject 1002 still demonstrated significantly less pain on XEN402

compared with placebo. The total amount of pain (combined AUC_{0-2} for PINRS) for the 2 hours after each induction period on active was 79% of placebo (95% CI = 67%, 92%; $P = .011$) (Fig. 1).

3.4. Subject 1003

The pain in subject 1003 before induction (basal pain) was lower on XEN402 when compared with placebo. This was particularly evident on the second day of dosing (Fig. 3B). The use of a heater was sufficient to consistently induce a severe pain response. In addition to the generally lowered basal pain while on XEN402, this subject also showed a significantly reduced total amount of pain (combined AUC_{0-2} for PINRS) for the 2 hours after each induction on XEN402 compared with placebo (Pain on XEN402 is 67% of placebo; 95% CI = 55%, 83%; $P = .004$) (Fig. 1). Furthermore, while on XEN402, the subject was able to tolerate the heater for longer, 34% greater than placebo (95% CI = -15%, 110%; $P = .158$), and the pain induced while on XEN402 returned more quickly to baseline levels, when compared with the placebo arm. The robust efficacy observed was clinically meaningful.

3.5. Subject 1004

In subject 1004, pain before induction (basal pain) was lower when on XEN402 compared with placebo (Fig. 3C). This subject was also responsive to the heat induction method. The total amount of pain (combined AUC_{0-2} of the PINRS) for the 2 hours after the induction was significantly diminished while on XEN402 when compared with placebo (pain on XEN402 was 12% of placebo; 95% CI = 2%, 75%; $P = .031$) (Fig. 1). Furthermore, when on XEN402, it took significantly longer to induce pain than on placebo (time to induction on active was 106% greater than on placebo (95% CI = 56%, 174%; $P < .001$). For some of the induction periods, heating was halted, as there was a failure to induce the pain while on XEN402. In addition, when pain was induced on XEN402, it was of shorter duration, returning rapidly to baseline levels when compared with that observed with placebo (Fig. 3C).

Subject 1004 showed greater improvements in pain suppression on day 2 of the XEN402 treatment when compared with the first day of dosing. On the morning of day 2 the investigator was completely unable to induce pain, despite a ~2-fold longer pain induction period, whereas there was robust pain induction during

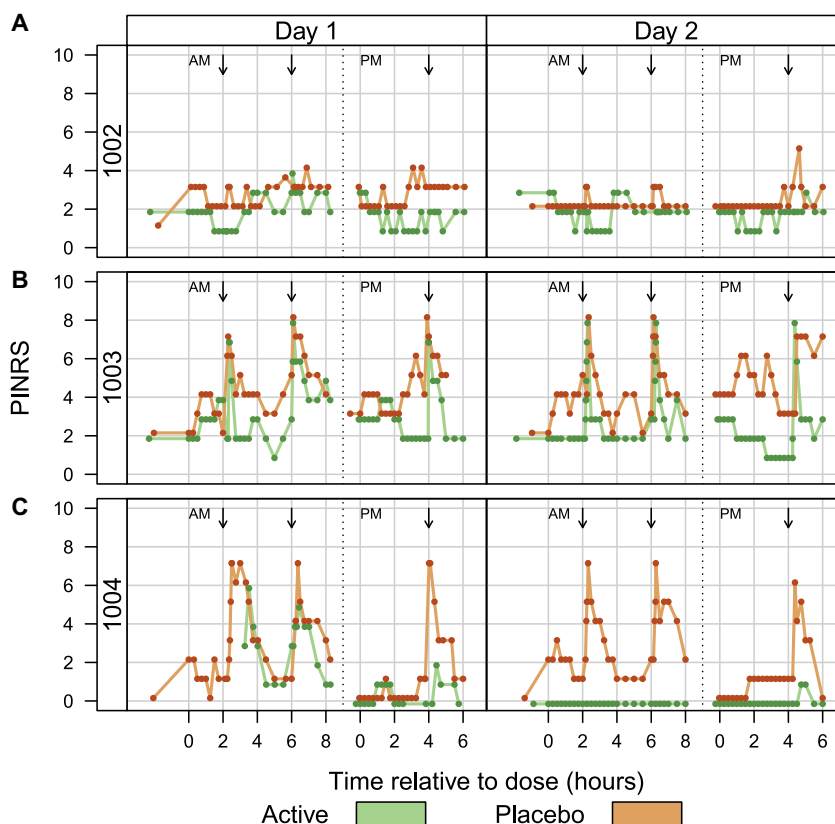


Fig. 3. PINRS scores for subjects 1002, 1003, and 1004. Pain intensity was scored by the subject throughout the day; each score is represented by an individual dot. (A) PINRS scores for subject 1002. (B) PINRS scores for subject 1003. (C) PINRS scores for subject 1004. Arrows represent the start time of induction at 2 and 6 hours after the morning dose and at 4 hours after the evening dose. For subject 1004, the morning induction was delayed on the active arm. AM = morning; PINRS = Pain Intensity Rating Scale (0 = no pain and 10 = worst pain imaginable); PM = evening; Time 0 = dose time.

the corresponding Day 2 period on placebo (up to 7 on PINRS). Subject 1004 reported to the investigator that he was able to take a hot shower for the first time in his life without any pain. Efficacy was clinically meaningful and provided robust evidence that XEN402 can pharmacologically inhibit $\text{Na}_v1.7$ mediated pain in this subject.

It is noteworthy that if the induction time had been standardized (ie, the same duration on each occasion), very limited induction of pain would have occurred during the active treatment period for subjects 1003 and 1004.

3.6. Adverse events

No serious adverse events (SAEs) were reported, and no AEs led to study discontinuation. The frequency and severity of AEs observed after active treatment was greater than after placebo treatment. Subjects 1002 and 1003 experienced dizziness and somnolence (predominantly transient and C_{max} related; data not shown), respectively, which led to reduction of their final administered dose of XEN402 on Day 2 of active dosing (from 400 mg to 300 mg in subject 1002 and to 200 mg in subject 1003). No clinically significant changes were observed in vital signs, clinical laboratory, physical examination, and ECG results during either treatment period.

4. Discussion

These data indicate that a novel sodium channel blocker is able to pharmacologically antagonize the effects of the mutant $\text{Na}_v1.7$ mediated IEM pain, and further demonstrate that such $\text{Na}_v1.7$

blockers may prove effective treatment options for the spontaneous debilitating pain associated with IEM. The data also suggest improved efficacy after repeat dosing which, if sustained, could lead to further reductions in pain with more chronic dosing.

Despite the small number of patients enrolled and the short dosing period (2 days), we were able to demonstrate reductions in pain measurements in all 4 subjects with statistically significant reductions in pain in all subjects with induced pain (subjects 1002, 1003, and 1004). After treatment with a novel sodium channel blocker, significant improvements were observed in the time taken to induce pain, the magnitude of induced pain, and/or reductions in the amount of pain experienced in the 2 hours after induction. This study supports the targeted use of genetic approaches to patient enrollment in clinical trials as an efficient way to interrogate pharmacological effects on the mechanism of interest and demonstrate clinically meaningful efficacy.

Because of the short treatment duration, it remains unknown whether subject 1001 (constant pain phenotype), would have experienced further improvements with more chronic dosing. The variable patient response to this drug could be due to differences in the causal missense mutations and their influences on drug binding kinetics, as well as the phenotypic variation and severity of IEM in this study. Factors influencing drug metabolism and pharmacokinetics might also explain differential drug responsiveness in different patients.

The results of this study should be interpreted with caution because of several important study limitations. These include the small sample size, its exploratory nature with a lack of prespecified endpoints or imputation methods, the short study duration, and

the heterogeneous study population. It is also possible that subjects could have become unblinded to active treatment because of the nature of CNS-related adverse events experienced. However, it is likely that knowledge of treatment would have had only a limited impact on the reported pain scores, given the robust nature of the pain induction methods used in each treatment period. Finally, in the majority of neuropathic pain syndromes, it is spontaneous pain rather than evoked that represents a major unmet medical need. Another limitation of this study, therefore, is that it deals primarily with evoked pain, for which the data were more compelling than those observed for spontaneous pain. Because of these limitations, no definitive conclusions or generalizations can be made. However, the data are consistent with evidence for target engagement and a probable analgesic effect, thereby providing sufficient evidence to warrant further investigations.

Here we have used a novel strategy for determining rapid proof-of-concept by demonstrating target engagement of the mutant target channel in humans. This approach may have broader utility serving as an early screening tool to reduce the high attrition rate and late stage failures in drug development.

Na_v1.7 genetic deficiency causes a broad human analgesia, whereby CIP patients are unable to experience pain after noxious stimuli. Here, by partially mimicking pharmacologically the Na_v1.7-loss-of-function CIP phenotype, this study showed that XEN402 is capable of blocking Na_v1.7-mediated pain associated with IEM. The expression and likely the function of Na_v1.7 are increased in certain nongenetic inflammatory and neuropathic pain conditions, including difficult-to-treat post-herpetic neuralgia and complex regional pain syndromes [1,16]. Thus, drugs capable of antagonizing Na_v1.7 may also be effective treatments not only for genetic disorders of enhanced Na_v1.7 function but for numerous painful conditions seen commonly in the general population. Such drugs would be a welcome addition to the treatment arsenal for both acute and chronic pain conditions, as significant numbers of patients being treated today are underresponding or failing to respond to available therapies.

Conflict of interest statement

M.H.L. was reimbursed by Xenon Pharmaceuticals for her role as investigator. J.P. was reimbursed for his role as statistician on this study. Y.P.G., N.P., R.N., C.J.C., C.W., C.E.Y., H.V., R.S., S.N.P., are employees and MRH is a consultant to Xenon Pharmaceuticals.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.pain.2011.09.008.

References

- [1] Beneng K, Renton T, Yilmaz Z, Yiangou Y, Anand P. Sodium channel Na v 1.7 immunoreactivity in painful human dental pulp and burning mouth syndrome. *BMC Neurosci* 2010;11:71.
- [2] Catterall WA, Dib-Hajj S, Meisler MH, Pietrobon D. Inherited neuronal ion channelopathies: new windows on complex neurological diseases. *J Neurosci* 2008;28:11768–77.
- [3] Cox JJ, Reimann F, Nicholas AK, Thornton G, Roberts E, Springell K, Karbani G, Jafari H, Mannan J, Raashid Y, Al-Gazali L, Hamamy H, Valente EM, Gorman S, Williams R, McHale DP, Wood JN, Gribble FM, Woods CG. An SCN9A channelopathy causes congenital inability to experience pain. *Nature* 2006;444:894–8.
- [4] Danziger N, Willer JC. Congenital insensitivity to pain. *Rev Neurol (Paris)* 2009;165:129–36.
- [5] Dib-Hajj SD, Rush AM, Cummins TR, Hisama FM, Novella S, Tyrrell L, Marshall L, Waxman SG. Gain-of-function mutation in Nav1.7 in familial erythromelalgia induces bursting of sensory neurons. *Brain* 2005;128:1847–54.
- [6] Drenth JP, Waxman SG. Mutations in sodium-channel gene SCN9A cause a spectrum of human genetic pain disorders. *J Clin Invest* 2007;117:3603–9.
- [7] Dworkin RH, Turk DC, Wyrwich KW, Beaton D, Cleeland CS, Farrar JT, Haythornthwaite JA, Jensen MP, Kerns RD, Ader DN, Brandenburg N, Burke LB, Cella D, Chandler J, Cowan P, Dimitrova R, Dionne R, Hertz S, Jadad AR, Katz NP, Kehlet H, Kramer LD, Manning DC, McCormick C, McDermott MP, McQuay HJ, Patel S, Porter L, Quessy S, Rappaport BA, Rauschkolb C, Revicki DA, Rothman M, Schmader KE, Stacey BR, Stauffer JW, Von Stein T, White RE, Witter J, Zavisic S. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *J Pain* 2008;9:105–21.
- [8] Farrar JT, Portenoy RK, Berlin JA, Kinman JL, Strom BL. Defining the clinically important difference in pain outcome measures. *Pain* 2000;88:287–94.
- [9] Fischer TZ, Gilmore ES, Estacion M, Eastman E, Taylor S, Melanson M, Dib-Hajj SD, Waxman SG. A novel Nav1.7 mutation producing carbamazepine-responsive erythromelalgia. *Ann Neurol* 2009;65:733–41.
- [10] Fischer TZ, Waxman SG. Familial pain syndromes from mutations of the Nav1.7 sodium channel. *Ann NY Acad Sci* 2010;1184:196–207.
- [11] Goldberg YP, MacFarlane J, MacDonald ML, Thompson J, Dube MP, Mattice M, Fraser R, Young C, Hossain S, Pape T, Payne B, Radomski C, Donaldson G, Ives E, Cox J, Younghusband HB, Green R, Duff A, Boltshauser E, Grinspan GA, Dimon JH, Sibley BG, Andria G, Toscano E, Kerdraon J, Bowsher D, Pimstone SN, Samuels ME, Sherrington R, Hayden MR. Loss-of-function mutations in the Nav1.7 gene underlie congenital indifference to pain in multiple human populations. *Clin Genet* 2007;71:311–9.
- [12] Lampert A, O'Reilly AO, Reeh P, Leffler A. Sodium channelopathies and pain. *Pflugers Arch* 2010;460:249–63.
- [13] Reimann F, Cox JJ, Belfer I, Diatchenko L, Zaykin DV, McHale DP, Drenth JP, Dai F, Wheeler J, Sanders F, Wood L, Wu TX, Karppinen J, Nikolajsen L, Mannikko M, Max MB, Kiselycznyk C, Poddar M, Te Morsche RH, Smith S, Gibson D, Kelempisioti A, Maixner W, Gribble FM, Woods CG. Pain perception is altered by a nucleotide polymorphism in SCN9A. *Proc Natl Acad Sci USA* 2010;107:5148–53.
- [14] Waxman SG. Nav1.7, its mutations, and the syndromes that they cause. *Neurology* 2007;69:505–7.
- [15] Yang Y, Wang Y, Li S, Xu Z, Li H, Ma L, Fan J, Bu D, Liu B, Fan Z, Wu G, Jin J, Ding B, Zhu X, Shen Y. Mutations in SCN9A, encoding a sodium channel alpha subunit, in patients with primary erythromelalgia. *J Med Genet* 2004;41:171–4.
- [16] Zhao P, Barr TP, Hou Q, Dib-Hajj SD, Black JA, Albrecht PJ, Petersen K, Eisenberg E, Wymer JP, Rice FL, Waxman SG. Voltage-gated sodium channel expression in rat and human epidermal keratinocytes: evidence for a role in pain. *Pain* 2008;139:90–105.